FOOD & ENVIRONMENT SAFETY ANALYTICAL LAB

Sample ID: 47440821-15
Date Issued: 8/26/24
Batch Result: Pass

Certificate of Analysis

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Grape Gas

Client: Pure Leaf Distribution

Total CBD	**ND**
Total THC	**27.11 %**
Total Cannabinoids	**30.88 %**

Analysis Summary

Residual Pesticides	Pass
Mycotoxins	Pass
Heavy Metals	Pass
Microbial Impurities	Pass

Sample Name:
Grape Gas

Batch Number:
PLD82224GG

Matrix:
Plant

Unit Mass:
1 g per unit

Sample ID:
47440821-15

Date Received:
8/21/2024

Approved By:
Marie True, M.S.
Laboratory Manager

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References: limit of detection (LOD), limit of quantitation (LOQ), not detected (ND), not tested (NT)

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Cannabinoid Analysis

Complete

Analyte	LOD (%)	LOQ (%)	Mass (%)	Mass (mg/g)
CBDV	0.0035	0.011	ND	ND
CBD	0.0030	0.0090	ND	ND
CBG	0.0038	0.011	ND	ND
CBDA	0.0017	0.0052	ND	ND
CBN	0.00080	0.0024	ND	ND
Delta 9-THC	**0.0022**	**0.0067**	**0.197**	**1.97**
Delta 8-THC	0.0020	0.0059	ND	ND
CBC	0.00070	0.0021	ND	ND
THCA	**0.0024**	**0.0073**	**30.687**	**306.87**
Total CBD			ND	ND
Total THC			**27.11**	**271.10**
Total Cannabinoids			**30.88**	**308.84**

Date Tested: 8/22/2024

Total THC = THCa * 0.877 + d9-THC + d8-THC

Total CBD = CBDa * 0.877 + CBD

Pesticide Analysis

Pass

Analyte	LOQ (ppm)	Limit (ppm)	Mass (ppm)	Status
Abamectin	0.050	0.10	ND	Pass
Acephate	0.050	0.10	ND	Pass
Acequinocyl	0.050	0.10	ND	Pass
Acetamiprid	0.050	0.10	ND	Pass
Aldicarb	0.050	0.00	ND	Pass
Azoxystrobin	0.050	0.10	ND	Pass
Bifenazate	0.050	0.10	ND	Pass
Bifenthrin	0.050	3.00	ND	Pass
Boscalid	0.050	0.10	ND	Pass
Captan	0.050	0.70	ND	Pass
Carbaryl	0.050	0.50	ND	Pass
Carbofuran	0.050	0.00	ND	Pass
Chlorantraniliprole	0.050	10.00	ND	Pass
Chlordane	0.050	0.00	ND	Pass
Chlorfenapyr	0.050	0.00	ND	Pass
Chlorpyrifos	0.050	0.00	ND	Pass
Clofentezine	0.050	0.10	ND	Pass
Coumaphos	0.050	0.00	ND	Pass
Cyfluthrin	0.050	2.00	ND	Pass
Cypermethrin	0.050	1.00	ND	Pass
Daminozide	0.050	0.00	ND	Pass
DDVP	0.050	0.00	ND	Pass
Diazinon	0.050	0.10	ND	Pass
Dimethoate	0.050	0.00	ND	Pass
Dimethomorph	0.050	2.00	ND	Pass
Ethoprophos	0.050	0.00	ND	Pass
Etofenprox	0.050	0.00	ND	Pass
Etoxazole	0.050	0.10	ND	Pass
Fenhexamid	0.050	0.10	ND	Pass
Fenoxycarb	0.050	0.00	ND	Pass
Fenpyroximate	0.050	0.10	ND	Pass
Fipronil	0.050	0.00	ND	Pass
Flonicamid	0.050	0.10	ND	Pass
Fludioxonil	0.050	0.10	ND	Pass

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Pesticide Analysis

Pass

Analyte	LOQ (ppm)	Limit (ppm)	Mass (ppm)	Status
Hexythiazox	0.050	0.10	ND	Pass
Imazalil	0.050	0.00	ND	Pass
Imidacloprid	0.050	5.00	ND	Pass
Kresoxim Methyl	0.050	0.10	ND	Pass
Malathion	0.050	0.50	ND	Pass
Metalaxyl	0.050	2.00	ND	Pass
Methiocarb	0.050	0.00	ND	Pass
Methomyl	0.050	1.00	ND	Pass
Methyl Parathion	0.050	0.00	ND	Pass
Mevinphos	0.050	0.00	ND	Pass
Myclobutanil	0.050	0.10	ND	Pass
Naled	0.050	0.10	ND	Pass
Oxamyl	0.050	0.50	ND	Pass
Paclobutrazol	0.050	0.00	ND	Pass
Pentachloronitrobenzene	0.050	0.10	ND	Pass
Permethrin	0.050	0.50	ND	Pass
Phosmet	0.050	0.10	ND	Pass
Piperonyl Butoxide	0.050	3.00	ND	Pass
Prallethrin	0.050	0.10	ND	Pass
Propiconazole	0.050	0.10	ND	Pass
Propoxur	0.050	0.00	ND	Pass
Pyrethrins	0.050	0.50	ND	Pass
Pyridaben	0.050	0.10	ND	Pass
Spinetoram	0.050	0.10	ND	Pass
Spinosad	0.050	0.10	ND	Pass
Spiromesifen	0.050	0.10	ND	Pass
Spirotetramat	0.050	0.10	ND	Pass
Spiroxamine	0.050	0.00	ND	Pass
Tebuconazole	0.050	0.10	ND	Pass
Thiacloprid	0.050	0.00	ND	Pass
Thiamethoxam	0.050	5.00	ND	Pass
Trifloxystrobin	0.050	0.10	ND	Pass

Date Tested: 8/22/2024

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Mycotoxins

Pass

Analyte	LOQ (µg/g)	Limit (µg/g)	Mass (µg/g)	Status
Aflatoxin B1	0.02	0.02	ND	Pass
Aflatoxin B2	0.02	0.02	ND	Pass
Aflatoxin G1	0.02	0.02	ND	Pass
Aflatoxin G2	0.02	0.02	ND	Pass
Ochratoxin A	0.02	0.02	ND	Pass

Date Tested: 8/22/2024

Heavy Metals Analysis

Pass

Analyte	LOQ (µg/g)	Limit (µg/g)	Mass (µg/g)	Status
Arsenic	0.050	0.200	ND	Pass
Cadmium	0.050	0.200	ND	Pass
Lead	0.125	0.500	0.163	Pass
Mercury	0.025	0.100	ND	Pass

Date Tested: 8/23/2024

Microbial Analysis

Pass

Test	Result (CFU/g)	Status
Aspergillus flavus	Absent / 1g	Pass
Aspergillus fumigatus	Absent / 1g	Pass
Aspergillus niger	Absent / 1g	Pass
Aspergillus terreus	Absent / 1g	Pass
Shiga-toxin producing *Escherichia coli*	Absent / 1g	Pass
Salmonella	Absent / 1g	Pass

Date Tested: 8/23/2024

CFU = Colony Forming Units

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Method References:	Testing Location
Cannabinoid Profile (UNODC)	FESA Labs - Santa Ana, CA
Multi-Residue Pesticide Analysis - (AOAC_200701)	FESA Labs - Santa Ana, CA
Mycotoxins Analysis - 5 compounds (FDA_MYC)	FESA Labs - Santa Ana, CA
Heavy Metals Analysis - 4 elements (EPA_200.8)	FESA Labs - Santa Ana, CA
Microbial Analysis - (FDABAM_4A_5_18)	FESA Labs - Santa Ana, CA

Official Methods of Analysis, Method 2018.11.AOAC INTERNATIONAL (modified), Lukas Vaclavik, Frantisek Benes, Alex Krmela, Veronika Svobodova, Jana Hajsolva, and Katerina Mastovska, "Quantification of Cannabinoids in Cannabis Dried Plant Materials, Concentrates, and Oils Liquid Chromatography-Diode Array Detection Technique with Optional Mass Spectrometric Detection," First Action Method, Journal of AOAC International, Future Issue

United Nations Office on Drugs and Crime - Recommended methods for identification and analysis of cannabis and cannabis products

Official Methods of Analysis, AOAC Official Method 2007.01, Pesticide Residues in Foods by Acetonitrile Extraction and Partitioning with Magnesium Sulfate, AOAC INTERNATIONAL (modified).

CEN Standard Method EN 15662: Food of plant origin - Determination of pesticide residues using GC-MS and/or LC-MS/MS following acetonitrile extraction/partitioning and clean-up by dispersive SPE - QuEChERS method.

Determination of Mycotoxins in Corn, Peanut Butter and Wheat Flour Using Stable Isotope Dilution Assay (SIDA) and Liquid Chromatography-Tandem Mass Spectrometry (LC-MS/MS) (modified).

Methods for the Determination of Metals in Environmental Standards - Supplement 1, EPA-600/R-94-111, May 1994.

"Determination of Metals and Trace Elements in Water and Wastes by Inductively Coupled Plasma-Mass Spectrometry", USEPA Method 200.8, Revision 5.1, EMMC Version (modified).

U.S. Food and Drug Administration, Bacteriological Analytical Manual, Chapter 4A, Diarrheagenic Escherichia coli; Chapter 5, Salmonella; Chapter 18, Yeasts, Molds and Mycotoxins (modified).

Testing Location:

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